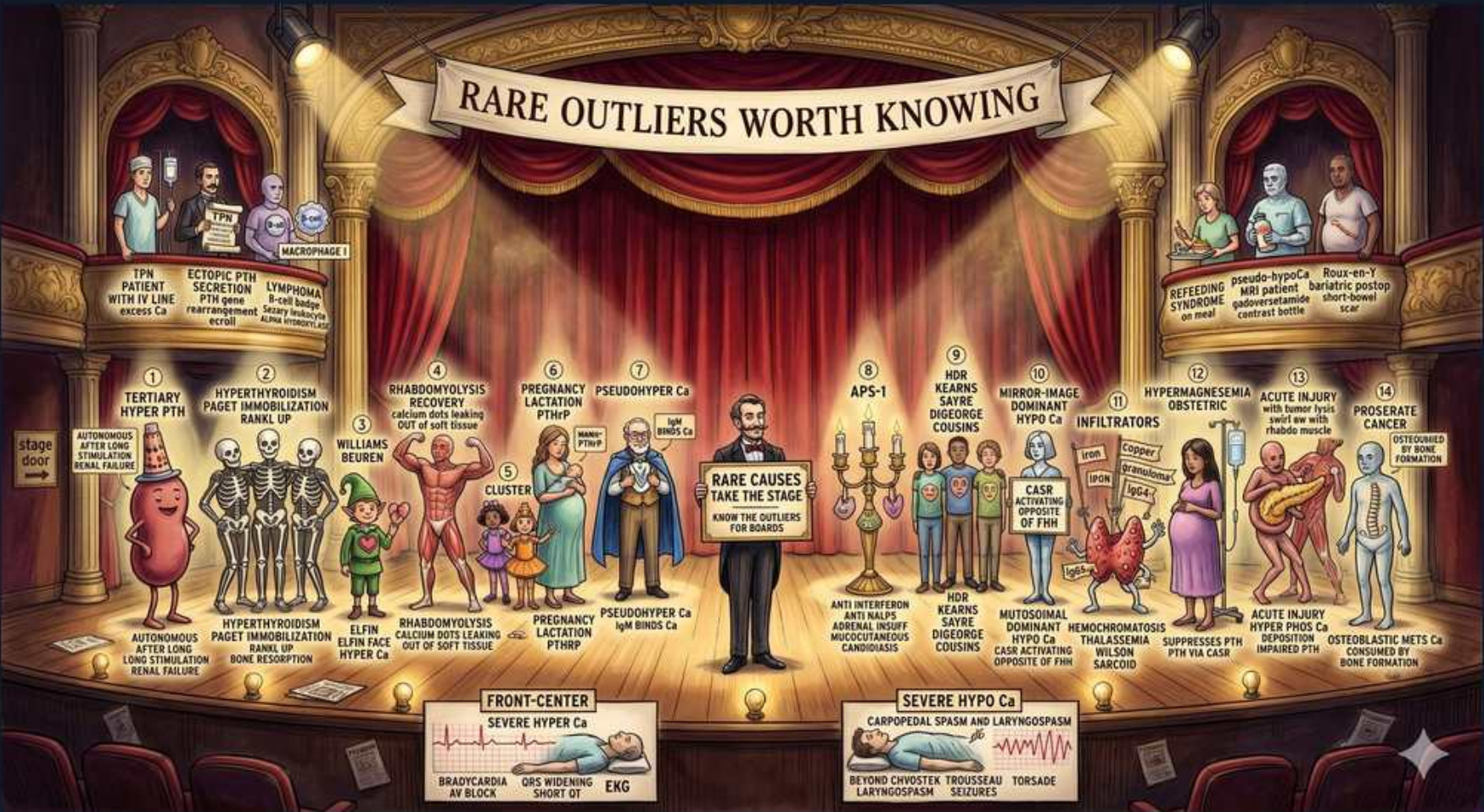


Calcium (Renal) — Rare Causes



1. Rare outliers worth knowing

WHAT YOU SEE

Banner: RARE OUTLIERS WORTH KNOWING

WHAT IT ENCODES

The board-edge causes beyond the common primary-hyperPTH-vs-malignancy split. These show up as 'second-order' questions: the labs don't fit the usual pattern, or a syndrome/context is the giveaway. The discriminator is almost always the combination of PTH, phosphate, urine calcium, and the clinical setting (CKD, pregnancy, autoimmune, mitochondrial, metastatic).

BOARD TRAP

WRONG: forcing every hypercalcemia into 'hyperPTH or PTHrP.' When labs are discordant (e.g., high PTH after a transplant, low Ca with prostate cancer), recognize a rare cause rather than over-anchoring on the common two.

2. Tertiary hyperPTH

WHAT YOU SEE

Kidney figure left, tertiary hyper PTH, renal failure

WHAT IT ENCODES

Tertiary hyperparathyroidism: after years of CKD-driven SECONDARY hyperPTH, the glands become AUTONOMOUS (hyperplastic) and keep over-secreting PTH even after the stimulus is gone — classically post renal transplant → high Ca, HIGH PTH, often low/normal phosphate after transplant. Treat with cinacalcet or parathyroidectomy.

BOARD TRAP

WRONG: calling persistent high PTH + high calcium after a kidney transplant 'primary' hyperPTH. The history of long-standing CKD/secondary hyperPTH that became autonomous defines TERTIARY disease.

3. Ectopic PTH secretion

WHAT YOU SEE

Stage act #2: a tumor figure labelled ECTOPIC PTH SECRETION / PTH GENE REARRANGEMENT — a non-parathyroid tumor literally making true PTH (high PTH, no gland source).

WHAT IT ENCODES

TRUE ectopic PTH secretion (intact PTH made by a non-parathyroid tumor — e.g., ovarian, lung, neuroendocrine) is exceedingly rare: high Ca with HIGH PTH but no parathyroid source on imaging. Far rarer than PTHrP-mediated humoral hypercalcemia.

BOARD TRAP

WRONG: assuming high PTH in a cancer patient means ectopic PTH. The overwhelmingly common malignancy mediator is PTHrP (which SUPPRESSES native PTH) — true ectopic PTH is a curiosity, not the default.

4. Paget / immobilization

WHAT YOU SEE

Skeleton, Paget, immobilization, hyperPTH bone

WHAT IT ENCODES

Paget disease of bone (disordered high-turnover remodeling, very high alk phos) is usually normocalcemic but causes hypercalcemia when high bone turnover meets IMMOBILIZATION → skeletal calcium release. Bisphosphonates treat both the Paget activity and the hypercalcemia.

BOARD TRAP

WRONG: expecting hypercalcemia in active ambulatory Paget. Paget patients are typically normocalcemic; it is IMMOBILIZATION (or a fracture) superimposed on high turnover that liberates calcium.

5. Williams / elfin

WHAT YOU SEE

Elfin-face child, Williams syndrome

WHAT IT ENCODES

Williams syndrome (7q11.23 deletion, elastin/ELN): elfin facies, supravalvular AORTIC STENOSIS, intellectual disability with a notably friendly/'cocktail-party' personality, and IDIOPATHIC INFANTILE HYPERCALCEMIA. A pediatric/genetic vignette cause of high calcium.

BOARD TRAP

WRONG: missing the cardiac association. A friendly child with elfin facies and hypercalcemia has Williams — and the key complication to screen for is supravalvular aortic stenosis.

6. Rhabdomyolysis recovery

WHAT YOU SEE

Muscle figure, rhabdo recovery, Ca into soft tissue

WHAT IT ENCODES

Rhabdomyolysis: EARLY hypocalcemia (calcium deposits into damaged muscle and binds liberated phosphate), then REBOUND hypercalcemia during recovery as that calcium is mobilized back out of healing tissue. The phase determines the direction of the calcium derangement.

BOARD TRAP

WRONG: aggressively repleting calcium during the early hypocalcemic phase of rhabdo. Unless symptomatic, hold off — the calcium will rebound (sometimes to hypercalcemia) as muscle recovers; over-repletion worsens the rebound.

7. Pregnancy/lactation PTHrP

WHAT YOU SEE

Pregnant figure, pregnancy/lactation PTHrP

WHAT IT ENCODES

During pregnancy/lactation the breast and placenta produce PTHrP to mobilize calcium for the fetus/milk → can cause maternal hypercalcemia. Usually mild; consider it before attributing hypercalcemia in a lactating woman to a tumor.

BOARD TRAP

WRONG: launching a malignancy workup for mild hypercalcemia in a lactating woman. Physiologic breast/placental PTHrP is the benign explanation — context first.

8. Pseudohypercalcemia

WHAT YOU SEE

Stage act #7: a figure labelled PSEUDOHYPER Ca / IgM BINDS Ca — a paraprotein clamping calcium so TOTAL Ca reads high while ionized stays normal.

WHAT IT ENCODES

Pseudohypercalcemia: a high paraprotein (multiple MYELOMA, Waldenström) binds calcium, raising TOTAL calcium while IONIZED calcium is normal — the patient is not truly hypercalcemic. Hemoconcentration/dehydration with high albumin does the same.

BOARD TRAP

WRONG: treating an elevated total calcium in a myeloma patient as an emergency without checking ionized Ca. If ionized is normal, it's pseudohypercalcemia — but note myeloma ALSO causes TRUE hypercalcemia via lytic bone disease, so verify with ionized calcium.

9. APS-1

WHAT YOU SEE

Stage act #8: a row of figures labelled APS-1 / ANTI-INTERFERON / ANTI-NALP5 / ADRENAL INSUFF / MUCOCUTANEOUS CANDIDIASIS — the candidiasis + hypoparathyroidism + Addison triad.

WHAT IT ENCODES

Autoimmune polyglandular syndrome type 1 (APECED, AIRE gene): the triad of chronic mucocutaneous CANDIDIASIS, autoimmune HYPOPARATHYROIDISM (→ hypocalcemia, low PTH), and ADRENAL insufficiency (Addison). A genetic cause of acquired hypoparathyroidism in the young.

BOARD TRAP

WRONG: working up isolated hypocalcemia without the syndrome lens. Hypoparathyroidism plus candidiasis plus adrenal insufficiency = APS-1 — and missing the adrenal component risks fatal adrenal crisis.

10. Kearns-Sayre / mito

WHAT YOU SEE

Kearns-Sayre, mitochondrial disorders, hypoCa

WHAT IT ENCODES

Mitochondrial disorders (Kearns-Sayre: progressive external ophthalmoplegia, pigmentary retinopathy, cardiac conduction block before age 20) can include HYPOPARATHYROIDISM → hypocalcemia. A rare multisystem cause of low PTH/low Ca.

BOARD TRAP

WRONG: overlooking the cardiac conduction defect in Kearns-Sayre. The hypocalcemia is a feature, but the life-threatening association is heart BLOCK (may need a pacemaker).

11. Wilson / hemochromatosis infiltrate

WHAT YOU SEE

Mirror-image figures, infiltrative, hemochromatosis/Wilson/thalassemia

WHAT IT ENCODES

Infiltrative/deposition diseases (hemochromatosis and transfusional iron overload in thalassemia, Wilson copper, sarcoid, metastases) can DESTROY the parathyroid glands → hypoparathyroidism with hypocalcemia and low PTH. Consider in patients with iron/copper overload syndromes.

BOARD TRAP

WRONG: attributing endocrine failure in hemochromatosis only to diabetes/hypogonadism. Iron deposition can also wreck the parathyroids — check calcium/PTH in the multi-organ infiltrative picture.

12. HyperPTH obstetric

WHAT YOU SEE

Right pregnant figure, hyperPTH obstetric

WHAT IT ENCODES

Maternal PRIMARY hyperparathyroidism in pregnancy (rare): maternal hypercalcemia crosses the placenta and suppresses fetal parathyroids → NEONATAL HYPOCALCEMIA/tetany after birth. Maternal disease may also cause miscarriage/pre-eclampsia; surgery is sometimes done in the 2nd trimester.

BOARD TRAP

WRONG: ignoring the newborn. A neonate with hypocalcemic tetany should prompt evaluation of the MOTHER for hyperparathyroidism — the fetal parathyroids were chronically suppressed in utero.

13. Acute kidney injury

WHAT YOU SEE

Right-side figure labelled ACUTE INJURY / HYPER PHOS / IMPAIRED Ca — AKI retaining phosphate and dropping calcium in the oliguric phase, with a rebound later.

WHAT IT ENCODES

AKI shifts calcium by phase: OLIGURIC phase → hyperphosphatemia and hypocalcemia (phosphate retention, reduced 1,25-D); RECOVERY/diuretic phase or rhabdo-AKI → rebound HYPERcalcemia as deposited calcium re-mobilizes. Direction depends on the stage of injury.

BOARD TRAP

WRONG: over-treating early AKI hypocalcemia. As in rhabdo, calcium often rebounds in recovery — replete only for symptoms/severe levels to avoid recovery-phase hypercalcemia.

14. Prostate cancer osteoblastic

WHAT YOU SEE

Far-right blue bone figure labelled PROSTATE CANCER / OSTEOLASTIC METS / Ca CONSUMED BY BONE FORMATION — bone-building mets soaking up calcium → HYPOcalcemia.

WHAT IT ENCODES

OSTEOLASTIC metastases (classically prostate cancer, also some breast and small-cell lung) consume calcium to build (sclerotic) bone → HYPOcalcemia, the opposite of osteolytic mets. Look for sclerotic lesions on imaging and a high alk phos with low calcium.

BOARD TRAP

WRONG: assuming all bone metastases raise calcium. OSTEOLYTIC mets (breast, myeloma) cause HYPERcalcemia; OSTEOLASTIC mets (prostate) cause HYPOcalcemia — the lesion type flips the direction.

15. Severe hyper/hypo banners

WHAT YOU SEE

Bottom: severe hyperCa short QT / severe hypoCa Chvostek Trousseau torsades

WHAT IT ENCODES

The extremes to recognize instantly: SEVERE HYPERcalcemia → bradyarrhythmia, SHORT QT, lethargy/coma, profound dehydration; SEVERE HYPOcalcemia → carpopedal spasm, LARYNGOSPASM, seizures, LONG QT/torsades. Both are emergencies needing immediate, opposite interventions.

BOARD TRAP

WRONG: swapping the QT findings. SHORT QT = HYPER-calcemia; LONG QT = HYPO-calcemia. This single mix-up is one of the most common ECG distractors on calcium questions.
